

Project Executable Files

Date	29 September 2026
Team ID	Team 10
Project Name	AI StudyBuddy
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes (package.json)
4	Database schema / seed files (if applicable)	Yes (Mongoose schemas in src/models/)
5	Environment configuration file (.env.example or similar)	Yes (.env with placeholder values)
6	Deployed application URL (if hosted)	NA (local development server)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes (Postman test collection in NM-StudyBuddy folder)
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

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ai-studybuddy/
├── .env                ← Environment variables (secrets)
├── .gitignore          ← Files excluded from Git
├── index.js            ← Entry point – Express server setup
├── package.json        ← Dependencies and scripts
├── uploads/            ← Temporary file upload storage
└── src/
    ├── controllers/
    │   ├── authController.js    ← Register, login, refresh, logout
    │   ├── materialController.js ← Upload, CRUD, AI features
    │   ├── aiController.js      ← Standalone AI endpoints
    │   └── adminController.js    ← Admin-only operations
    ├── middleware/
    │   ├── auth.js              ← JWT verification + admin check
    │   └── upload.js            ← Multer file upload config
    ├── models/
    │   ├── User.js              ← User schema + password hashing
    │   └── Material.js          ← Material schema + AI fields
    ├── routes/
    │   ├── auth.js              ← /api/auth/* routes
    │   └── materials.js          ← /api/materials/* routes
  
```

		ai.js	← /api/ai/* routes
		admin.js	← /api/admin/* routes
		utils/	
		db.js	← MongoDB connection
		gemini.js	← Google Gemini AI integration
		tokens.js	← JWT token generation

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://localhost:5000 (local development server)
Login Credentials (Demo)	Register a new user via POST /api/auth/register with any name, email, password. For admin access, register with role: "admin".
Platform / Hosting Provider	Local Node.js server + MongoDB Atlas (cloud database)
Repository Link	(Paste your GitHub repository link here)
Demo Video Link	(Paste your Google Drive demo video link here)

Step 4: Run Instructions

- Clone the repository: git clone <repository-url>
- Navigate to the project folder: cd ai-studybuddy
- Install dependencies: npm install
- Create a .env file in the root directory with the following variables: PORT=5000, MONGO_URI=your_mongodb_connection_string, JWT_ACCESS_SECRET=your_jwt_secret, JWT_REFRESH_SECRET=your_jwt_refresh_secret, GEMINI_API_KEY=your_gemini_api_key, NODE_ENV=development
- Start the development server: npm start
- Verify the server is running: Open http://localhost:5000 in browser or Postman — should return {"message":"AI StudyBuddy API is running"}
- Test the API using Postman with the endpoints documented in API_DOCUMENTATION.md

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	No frontend UI — backend API only	Use Postman to test all endpoints. Frontend planned as future enhancement.
2	AI generation time depends on Gemini API response	Typically 3-8 seconds. Retry if Gemini API is temporarily slow.
3	PDF text extraction is basic (plain text only, no OCR)	Use .txt or .md files for best results. Complex PDFs with images may not extract fully.